Statistical Signal Processing
Lecture 01:
State space representation of LTIC and LTID systems

Literature and Problems: See course schedule

Agenda (preliminary):

SLIDES:

- **intro to course STATSIG subjects**
- **practical information about course STATSIG**

LTIC: CONTINUOUS-TIME

External model (unique)

- **SISO LTIC system differential equation**
- **transfer function H(s), tf**

Internal model (not unique)

- **basic idea: from 1 n'th order diff.eq. → n first order diff.eq.**
- **nx1 state vector, x(t)**
- **dynamical SSC model (State Space Continuous), ss**
- **model matrices A,B,C,D**
- **block diagram**
- **SSC versus H(s) advantages**

Transformation from H(s) → SSC, tf2ss ss

- ***introduce dummy variable V(s), inverse Laplace U(s) and Y(s)***
- ***choose state vector, manipulate → SSC***

Transformation from SSC → H(s), ss2tf tf

- ***Laplace SSC and manipulate***

H(s) poles = A eigenvalues

State transformation

- ***E*z(t) = x(t)***
- ***modal representation (diagonalization of A), eig***

LTID: DISCRETE-TIME

External model (unique)

- **SISO LTID system difference equation**
- **transfer function H(z), tf**

Internal model (not unique)

- **basic idea: from 1 N'th order diff.eq. → N first order diff.eq.**
- **Nx1 state vector, x(n)**
- **dynamical SSD model (State Space Discrete), ss**
- **model matrices A,B,C,D**
- **block diagram**
- **SSD versus H(z) advantages**

Transformation from H(z) → SSD, tf2ss ss

- **introduce dummy variable V(z), inverse Z of U(z) and Y(z)**
- **choose state vector, manipulate → SSD**

Transformation from SSD → H(z), ss2tf tf

- **Z of SSD and manipulate**

H(z) poles = A eigenvalues

State transformation

- **E*z(n) = x(n)**
- **modal representation (diagonalization of A), eig**

Generalizations

- **time-varying systems**
- **non-linear systems**
- **MIMO systems**